

STM32 GPIO Control Registers

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EE2028

Unless stated otherwise, for the second part of the module,

‘STM32 Chip’ = STM32L4S5VIT6

‘Board’ = B-L4S5I-IOT01A

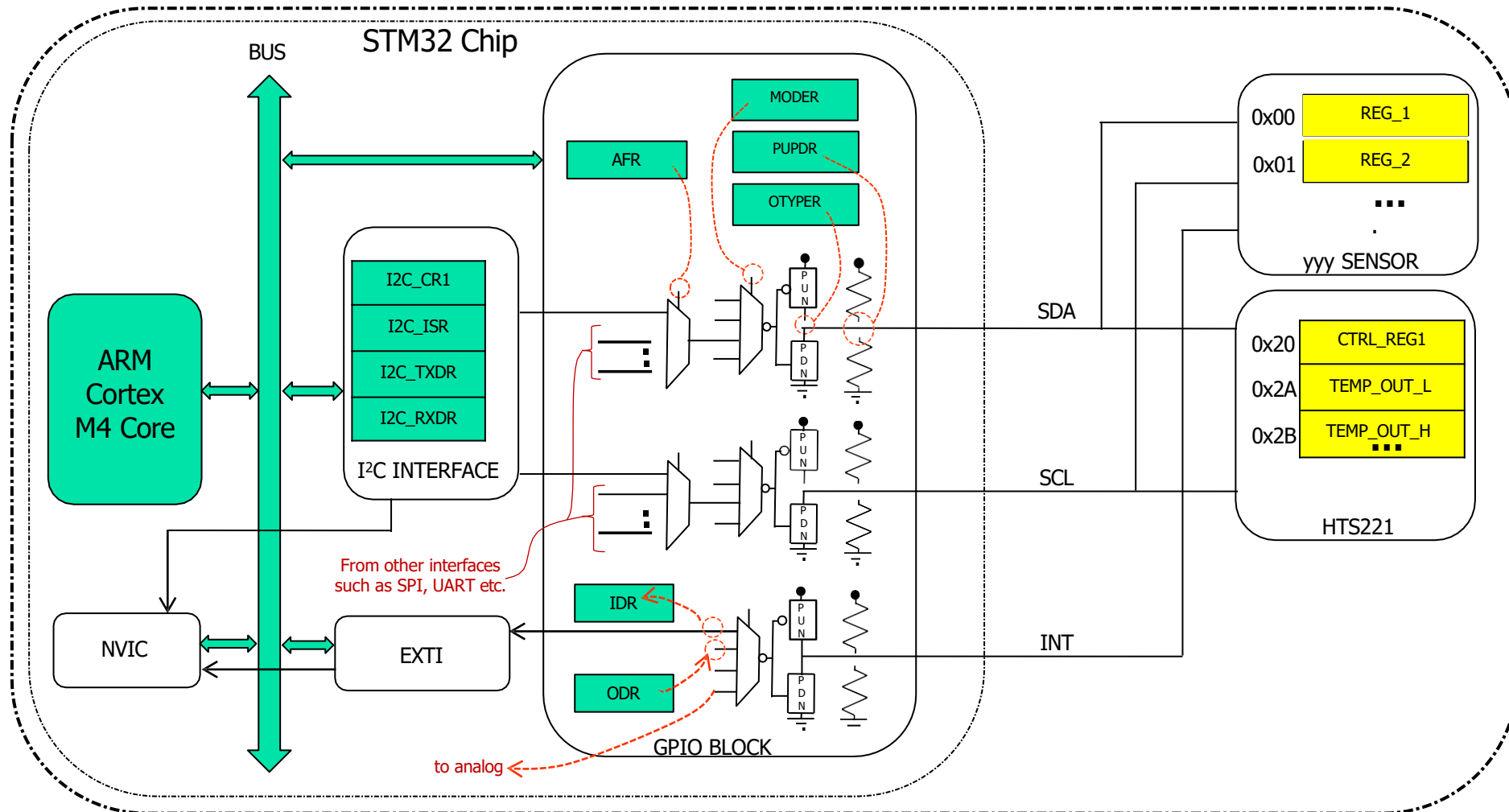
Acknowledgement: Slides adapted from Dr Rajesh Panicker



OVERVIEW

- GPIO Control Registers
 - MODER (Mode Register)
 - AFRL/H (Alternate Function Low/High Register)
 - PUPDR (Pull-Up Pull-Down Register)
 - OTYPER (Output Type Register)
- Open Drain Application

SYSTEM LEVEL OVERVIEW



GPIO CONTROL REGISTER - GPIOx_MODER

- MODER (Address Offset = 0x00) : Mode Register : First level of multiplexing
- GPIOx_MODER bits MODEy[1:0] configures the mode of pin y (y=15:0) of port x (x = A, B, C, D, E, F, G, H, I) to be
 - **00**: Input mode
 - **01**: General purpose output mode
 - **10**: Alternate function (AF) mode
 - **11**: Analog mode (reset state)
- 2 bits per pin -> 1 MODER register needed per *port* [port = up to 16 pins]

E.g. To configure Port B Pin 14 as GPIO output (x=B, y=14):

MODE14[1:0] of GPIOB_MODER, i.e.,

GPIOB_MODER [29:28] = **01**

GPIOB_MODER address =

AHB2 base (0x48000000) +

GPIOB offset (1 KB = 0x400) +

MODER offset (0x00) = 0x48000400

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODE15[1:0]		MODE14[1:0]		MODE13[1:0]		MODE12[1:0]		MODE11[1:0]		MODE10[1:0]		MODE9[1:0]		MODE8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODE7[1:0]		MODE6[1:0]		MODE5[1:0]		MODE4[1:0]		MODE3[1:0]		MODE2[1:0]		MODE1[1:0]		MODE0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

GPIO CONTROL REGISTER - GPIOx_AFRL/H

- AFRL/H (Address Offset = 0x20/0x24) : Alternate Function Low/ High Registers : Second level multiplexing : To choose which of the 16 internal peripherals/interfaces a pin is connected to, when GPIOx_MODER bits = 10 (alternate function)
- GPIOx_AFRL/H bits AFSELY[3:0]: Alternate function selection for port x pin y (y = 7:0 for AFRL; y = 15:8 for AFRH)

- 0000: AF0
- 0001: AF1
- 0010: AF2
-
- 1111: AF15

GPIOx_AFRL

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFSEL7[3:0]				AFSEL6[3:0]				AFSEL5[3:0]				AFSEL4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFSEL3[3:0]				AFSEL2[3:0]				AFSEL1[3:0]				AFSEL0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

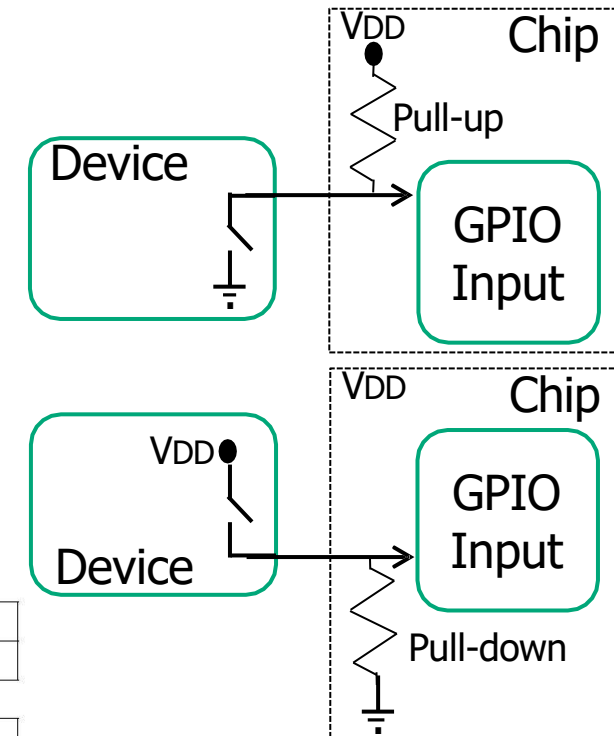
GPIOx_AFRH

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFSEL15[3:0]				AFSEL14[3:0]				AFSEL13[3:0]				AFSEL12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFSEL11[3:0]				AFSEL10[3:0]				AFSEL9[3:0]				AFSEL8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

- Since there are 16 alternate functions per pin, 4 AFSEL bits are required per pin. Hence, 2 AFR registers (AFRL and AFRH) per port
- E.g., To configure Port C Pin 12 as UART5_TX (x=C, y=12): AFSEL12[3:0] of GPIOC_AFRH, i.e., GPIOC_AFRH[19:16] = 4'b1000 (UART5_TX is AF8)

GPIO CONTROL REGISTER - GPIOx_PUPDR

- PUPDR (Address Offset = 0x0C) : Pull-up/pull-down register: Selects whether an internal pull-up / pull-down resistor is connected to a pin
- GPIOx_PUPDR bits PUPDy[1:0] configures pull-up/down setting of port x pin y (y=15:0)
 - 00**: No pull-up, pull-down: Used when the pin is not an input, or when we do not wish to set a default value to input. When we are confident that the external device will drive a strong LOW or HIGH and won't be floating
 - 01**: Pull-up: Sets the default input to HIGH. Used when the pin is an input which could be floating (unconnected / not driven actively by the external device at times). The default HIGH is overridden by the strong LOW driven by the external device
 - 10**: Pull-down: Similar to above, but sets the default input to LOW
 - 11**: Reserved: do not write 11 to PUPD bits
- 2 bits per pin -> 1 PUPDR register needed per port

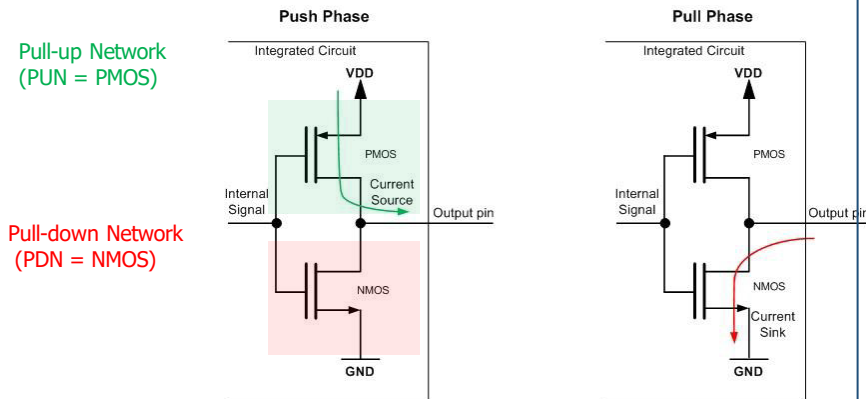


31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PUPD15[1:0]		PUPD14[1:0]		PUPD13[1:0]		PUPD12[1:0]		PUPD11[1:0]		PUPD10[1:0]		PUPD9[1:0]		PUPD8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PUPD7[1:0]		PUPD6[1:0]		PUPD5[1:0]		PUPD4[1:0]		PUPD3[1:0]		PUPD2[1:0]		PUPD1[1:0]		PUPD0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

GPIO CONTROL REGISTER - GPIOx_OTYPER

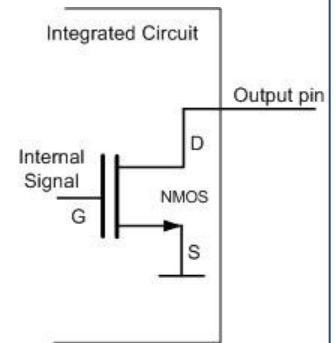
- OTYPER (Address Offset = 0x04) : Output Type Register
- GPIOx_OTYPER bits OT_y configures OTYPE of port x pin y (y=15:0)

0: Output push-pull (reset state) : Output is either a strong HIGH or a strong LOW. Cannot connect multiple outputs together



1: Output open-drain : Output is either a strong LOW or High-impedance (Z, also called floating, weak HIGH etc.)

PUN is disconnected. Allows multiple outputs to be connected together to one bus. Writing a logic 1 (weak HIGH) = disconnected from bus

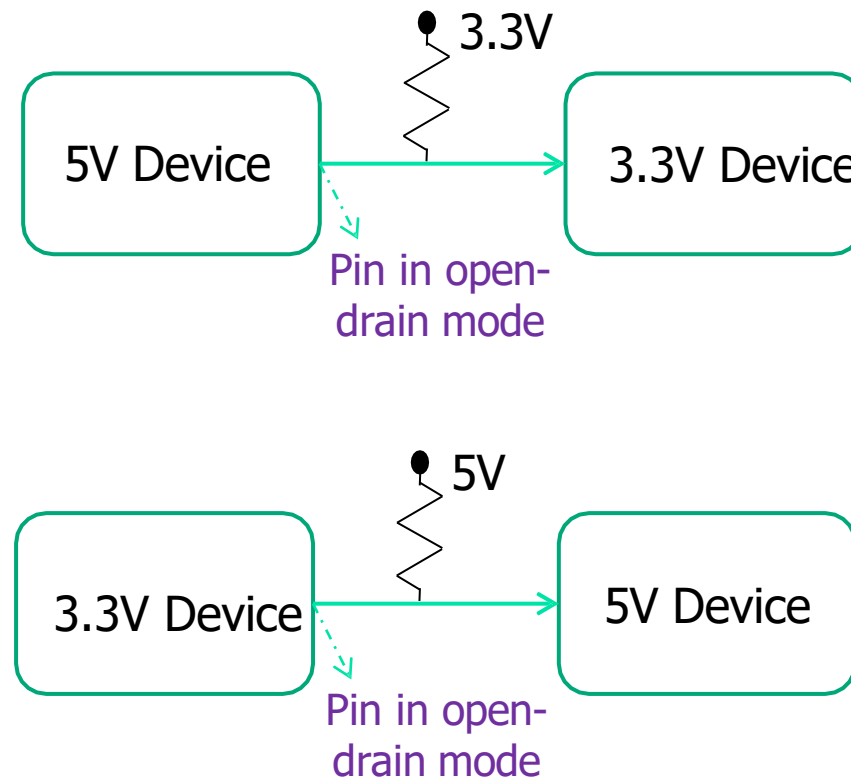


- 1 bit per pin: OTYPER register needed per port with 16 bits unused/reserved (must be kept at reset level)
- E.g. To configure Port E Pin 11 as Output push-pull (x=E, y=11): OT₁₄ of GPIOE_OTYPER, i.e., GPIOE_OTYPER[11] = 0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

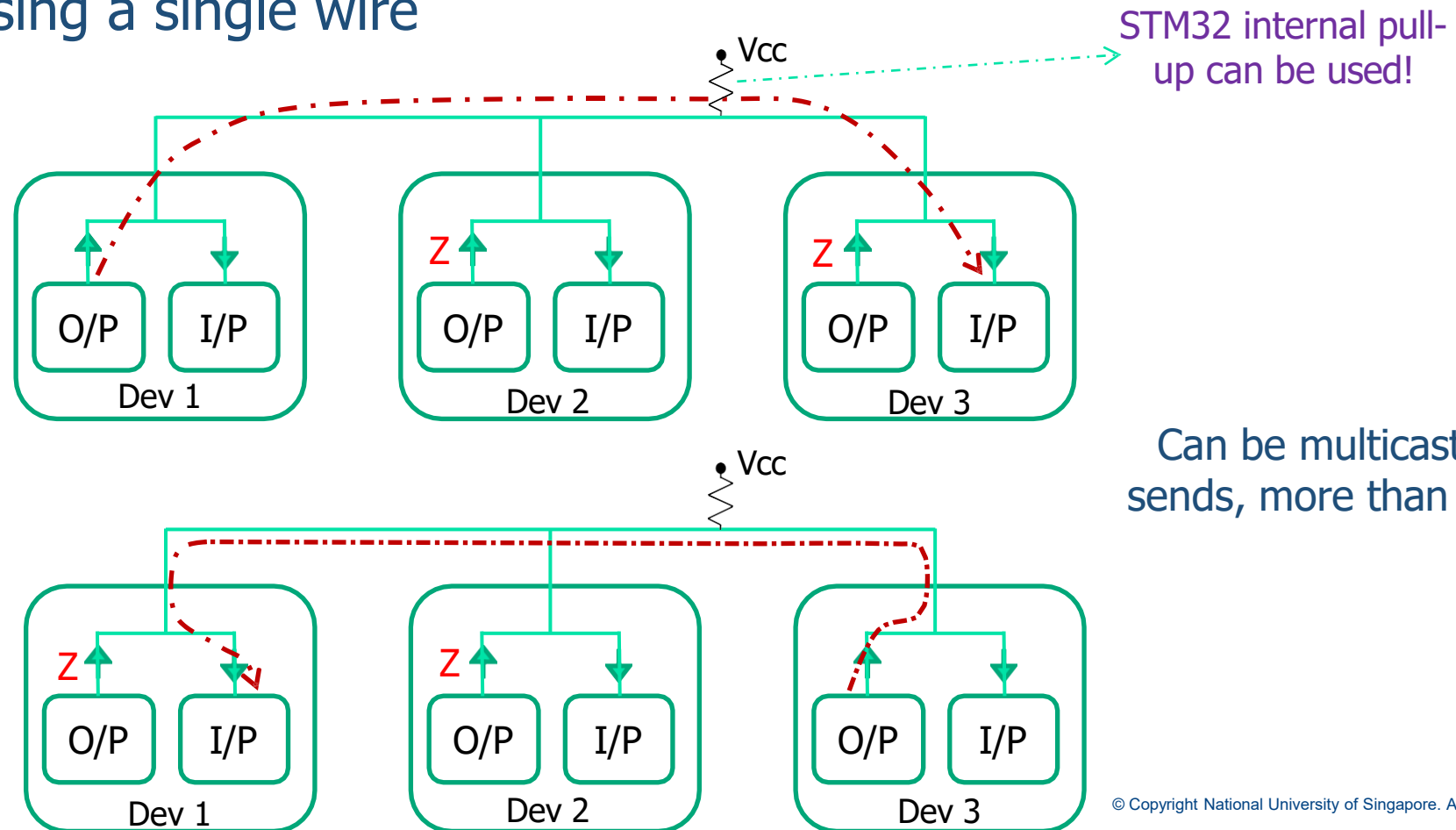
OPEN-DRAIN: INTERFACING VIA LEVEL-SHIFTING

- Open-drain mode can be used to interface between logic families using different voltage levels



OPEN-DRAIN: BIDIRECTIONAL COMMUNICATION (WIRED-AND)

- Open-drain mode allows bi-directional communication between multiple devices using a single wire



Can be multicast (one device sends, more than one receives)!

CONCLUSION

- GPIO Control Registers
 - MODER
 - AFRL/AFRH
 - PUPDR
 - OTYPER
- Open Drain Application
 - Interfacing via Level-shifting
 - Bidirectional communication